Ayush Kumar

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Summary

Final-Year B.Tech Information Technology student at IIIT Bhubaneswar specializing in Data Structures & Algorithms, Full-Stack MERN Engineering, and Applied AI/ML. Experienced in architecting scalable RESTful APIs, autonomous agent workflows (LangGraph), vector search pipelines (pgvector), and containerized microservices (Docker), targeting Software Development Engineer (SDE) Full-Time roles.

Education

International Institute of Information Technology (IIIT) — **Bhubaneswar, Odisha**
Bachelor of Technology in Information Technology (CGPA: 7.48) — *2023 – 2027*

Delhi Public School, Nalco Nagar — **Angul, Odisha**
Class XII (Senior Secondary): 88.6% | Class X (Secondary): 98.2% — *2020 – 2023*

Technical Skills

Languages: Java, JavaScript (ES6+), Python, TypeScript, SQL, HTML5, CSS3

Web Technologies: React.js, Node.js, Express.js, FastAPI, Flask, Vite, Tailwind CSS, Bootstrap 5, EJS

AI & Machine Learning: LangGraph, RAG Pipelines, pgvector, Scikit-learn, XGBoost, Pandas, NumPy, Google Gemini API

Databases & DevOps: PostgreSQL, MongoDB Atlas, Docker, Docker Compose, Git, GitHub Actions (CI/CD), Render, Cloudinary

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, RESTful APIs

Projects

Feasto | *React.js, Node.js, Express.js, MongoDB Atlas, Razorpay, Google Gemini API, Docker* GitHub | Live Demo

- Engineered a production-grade food delivery ecosystem following **MERN architecture** with decoupled customer and admin management portals, featuring real-time driver delivery tracking with **Leaflet GPS** and persistent cart state.
- Integrated **Razorpay Payment Gateway** with cryptographic **HMAC-SHA256** webhook signature verification, ensuring zero transaction discrepancies across shopping carts, order lifecycles, and database states.
- Embedded an AI food concierge using **Google Gemini API** for personalized dish recommendations; enforced **JWT role-based access control (RBAC)**, Cloudinary CDN image pipelines, and multi-container **Docker** deployment.

StayNest | *Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5, Cloudinary, Razorpay, PDFKit* GitHub | Live Demo

- Developed a full-stack accommodation marketplace adopting **MVC architecture** with full **CRUD** listings, geospatial search filters, verified customer reviews, and responsive UI built with **Bootstrap 5**.
- Designed a collision-proof booking engine featuring interactive blackout calendar scheduling, **Razorpay** payment verification, and programmatic server-side PDF invoice generation using **PDFKit**.
- Hardened backend application security with **Passport.js** session authentication, **Joi** input validation schemas, **Helmet** HTTP security headers, and **MongoDB Atlas** cloud storage.

Smart House Price Predictor | *Python, Flask, XGBoost, Scikit-learn, Pandas, NumPy, Docker* GitHub | Live Demo

- Trained a California real estate valuation engine achieving $R^2 = 83.93\%$ and **17.52% MAPE** ($30,245 MAE) using an optimized **XGBoost** gradient-boosted ensemble on 20,000+ property records.
- Formulated distribution-free **Split Conformal Prediction** delivering mathematically guaranteed **90% confidence intervals** ($\pm\$68,169$), eliminating financial tail risk compared to traditional point estimates.
- Deployed Explainable AI (**XAI**) feature attribution telemetry and defensive input validation, serving predictions via a low-latency (<20ms) **Flask REST API** containerized with **Docker** and continuous **GitHub Actions CI**.

AgentIQ | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, TypeScript, Docker Compose* GitHub | Live Demo

- Architected an enterprise agentic RAG platform using **FastAPI** and **LangGraph**, orchestrating bounded multi-step reasoning (≤ 4 steps) with deterministic fallbacks, execution timeouts, and sub-second retrieval latency.
- Pioneered hybrid semantic search via **PostgreSQL 16** and **pgvector** (HNSW indexing, 0.45 cosine cutoff) with source citations, protected by a 5-layer SQL guardrail with **sqlglot AST parsing** and **RBAC**.
- Constructed an enterprise-grade 3-pane UI in **React 18** and **TypeScript**, containerizing full-stack services with multi-stage **Docker** and establishing continuous **GitHub Actions CI/CD** testing.

Achievements & Coding Profiles

- Solved **200+ Data Structures & Algorithms** problems in **Java** across **LeetCode** and **GeeksforGeeks**, demonstrating proficiency in algorithmic complexity analysis, graph algorithms, and data-structure design.
- Containerized and deployed **4 production systems** (Feasto, StayNest, SmartHouse, AgentIQ) with **Docker**, **CI/CD** pipelines, and cloud hosting on **Render**.
- Demonstrated sustained academic excellence with **98.2% in Class X** and **88.6% in Class XII**; actively pursuing B.Tech in IT at **IIIT Bhubaneswar**.